Calculus Formula Sheet

Derivatives

Definition and Notation

If $y = f(x)$ then the derivative is defined to be

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

An equivalent definition of the derivative at $x = a$ is

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

If $y = f(x)$ then all of the following are equivalent notations for the derivative:

$$f'(x) = y' = \frac{df}{dx} = \frac{dy}{dx} = \frac{d}{dx}(f(x)) = Df(x)$$

If $y = f(x)$ then all of the following are equivalent notations for the derivative evaluated at $x = a$:

$$f'(a) = y'\big|_{x=a} = \frac{df}{dx}\Big|_{x=a} = \frac{dy}{dx}\Big|_{x=a} = Df(a)$$

Interpretation of the Derivative

If $y = f(x)$ then:

1. $m = f'(a)$ is the slope of the tangent line to $y = f(x)$ at $x = a$, and the equation of the tangent line at $x = a$ is given by

$$y = f(a) + f'(a)(x - a)$$

2. $f'(a)$ is the instantaneous rate of change of $f(x)$ at $x = a$.

3. If $f(t)$ is the position of an object at time t, then $f'(a)$ is the velocity of the object at $t = a$.

Behavior of the Accumulation Function

An accumulation function is of the form $F(x) = \int_a^x f(t)\,dt$, where a is a constant and f is continuous. Its derivative is $F'(x) = f(x)$.

$F(x)$ **is/has**	**when**	**or**
Increasing	$F'(x) > 0$	$f(x) > 0$
Decreasing	$F'(x) < 0$	$f(x) < 0$
Relative maximum	$F'(x)$ changes sign $+$ to $-$	$f(x)$ changes sign $+$ to $-$
Relative minimum	$F'(x)$ changes sign $-$ to $+$	$f(x)$ changes sign $-$ to $+$
Concave up	$F''(x) > 0$	$f'(x) > 0$
Concave down	$F''(x) < 0$	$f'(x) < 0$
Point of inflection	$F''(x) = 0$ or DNE, *and* F'' changes sign	$f'(x) = 0$ or DNE, *and* f' changes sign

Summation and Series Formulas

1. $\displaystyle\sum_{k=1}^{n} c = nc$
2. $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$
3. $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$
4. $\displaystyle\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$
5. $\displaystyle\sum_{k=1}^{\infty} r^k = \frac{r}{1-r} \quad (|r| < 1)$
6. $\displaystyle\sum_{k=0}^{\infty} r^k = \frac{1}{1-r} \quad (|r| < 1)$
7. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k} = \text{diverges}$
8. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$
9. $\displaystyle\sum_{k=1}^{n} (a_k \pm b_k) = \sum a_k \pm \sum b_k$
10. $\displaystyle\sum_{k=1}^{n} c\,a_k = c\sum a_k$
11. Geometric term: $a_n = ar^{n-1}$
12. Finite geometric sum: $S_n = \dfrac{a(1-r^n)}{1-r} \quad (r \neq 1)$
13. Infinite geometric sum: $S = \dfrac{a}{1-r} \quad (|r| < 1)$

Derivatives

1. $\dfrac{d}{dx}[x] = 1$
2. $\dfrac{d}{dx}[u \pm v] = u' \pm v'$
3. $\dfrac{d}{dx}[cv] = c\,v'$
4. $\dfrac{d}{dx}\left[\dfrac{u}{v}\right] = \dfrac{vu' - uv'}{v^2}$
5. $\dfrac{d}{dx}[c] = 0$
6. $\dfrac{d}{dx}[uv] = uv' + vu'$
7. $\dfrac{d}{dx}[x^n] = nx^{n-1}$ (any real n)
8. $\dfrac{d}{dx}[f(g(x))] = f'(g(x))\,g'(x)$
9. $\dfrac{d}{dx}[\ln|u|] = \dfrac{1}{u}\,u' \quad (u \neq 0)$
10. $\dfrac{d}{dx}[\ln(u)] = \dfrac{1}{u}\,u' \quad (u > 0)$
11. $\dfrac{d}{dx}[e^u] = e^u\,u'$
12. $\dfrac{d}{dx}[\log_a(u)] = \dfrac{1}{u\ln(a)}\,u' \quad (u > 0)$
13. $\dfrac{d}{dx}[a^u] = a^u(\ln a)\,u'$
14. $\dfrac{d}{dx}[\sin^{-1}(u)] = \dfrac{1}{\sqrt{1-u^2}}\,u'$
15. $\dfrac{d}{dx}[\cos^{-1}(u)] = \dfrac{-1}{\sqrt{1-u^2}}\,u'$
16. $\dfrac{d}{dx}[\tan^{-1}(u)] = \dfrac{1}{1+u^2}\,u'$
17. $\dfrac{d}{dx}[\cot^{-1}(u)] = \dfrac{-1}{1+u^2}\,u'$
18. $\dfrac{d}{dx}[\csc^{-1}(u)] = \dfrac{-1}{|u|\sqrt{u^2-1}}\,u'$
19. $\dfrac{d}{dx}[\sec^{-1}(u)] = \dfrac{1}{|u|\sqrt{u^2-1}}\,u'$
20. $\dfrac{d}{dx}[\sin u] = (\cos u)\,u'$
21. $\dfrac{d}{dx}[\cos u] = -(\sin u)\,u'$
22. $\dfrac{d}{dx}[\tan u] = (\sec^2 u)\,u'$
23. $\dfrac{d}{dx}[\cot u] = -(\csc^2 u)\,u'$
24. $\dfrac{d}{dx}[\sec u] = (\sec u\tan u)\,u'$
25. $\dfrac{d}{dx}[\csc u] = -(\csc u\cot u)\,u'$
26. $\dfrac{d}{dx}[\sinh u] = (\cosh u)\,u'$
27. $\dfrac{d}{dx}[\cosh u] = (\sinh u)\,u'$
28. $\dfrac{d}{dx}[\tanh u] = (\operatorname{sech}^2 u)\,u'$
29. $\dfrac{d}{dx}[\coth u] = -(\operatorname{csch}^2 u)\,u'$
30. $\dfrac{d}{dx}[\operatorname{sech} u] = -(\operatorname{sech} u\tanh u)\,u'$
31. $\dfrac{d}{dx}[\operatorname{csch} u] = -(\operatorname{csch} u\coth u)\,u'$

Domains: $\sin^{-1}, \cos^{-1}$: $|u| < 1$; $\sec^{-1}, \csc^{-1}$: $|u| > 1$; $\log_a u$: $u > 0$; $a^u, \log_a u$: $a > 0,\ a \neq 1$.

Special Limits

Angles are in **radians**. In every case the variable approaches the limit point without equaling it (so $x \neq 0$, or $f(x) \neq 0$ near $x = a$).

1. $\displaystyle\lim_{x\to 0} \frac{\sin x}{x} = 1$
2. $\displaystyle\lim_{x\to 0} \frac{x}{\sin x} = 1$
3. $\displaystyle\lim_{x\to a} \frac{\sin(f(x))}{f(x)} = 1$ (if $f(x) \to 0$)
4. $\displaystyle\lim_{x\to a} \frac{f(x)}{\sin(f(x))} = 1$ (if $f(x) \to 0$)
5. $\displaystyle\lim_{x\to 0} \frac{\tan x}{x} = 1$
6. $\displaystyle\lim_{x\to 0} \frac{1-\cos x}{x} = 0$
7. $\displaystyle\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$
8. $\displaystyle\lim_{x\to 0} \frac{x}{1-\cos x} = \text{DNE}$ ($-\infty$ from the left, $+\infty$ from the right)
9. $\displaystyle\lim_{x\to 0} \frac{e^x - 1}{x} = 1$
10. $\displaystyle\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$ ($1 + x > 0$)
11. $\displaystyle\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e$

Logarithm & Exponential Facts

1. $\ln(1) = 0$
2. $\ln(0)$ undefined; $\displaystyle\lim_{x\to 0^+} \ln x = -\infty$
3. $e^0 = 1$
4. $\ln(e^a) = a$
5. $e^{\ln(a)} = a \quad (a > 0)$

Notation & Differentiating Inverses

- $(g(x))^2 = g^2(x)$ (but g^{-1} means the inverse, not $1/g$)
- $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$ (needs $f'(f^{-1}(x)) \neq 0$)

Theorems, Continuity, and Curve Analysis

Continuity at $x = c$

1. $f(c)$ is defined.
2. $\lim_{x \to c} f(x)$ exists.
3. $\lim_{x \to c} f(x) = f(c)$.

Squeeze Theorem

If $g(x) \le f(x) \le h(x)$ for all x near a (except possibly at a), and $\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$, then $\lim_{x \to a} f(x) = L$.

Intermediate Value Theorem (IVT)

If f is continuous on $[a, b]$ and N is between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that $f(c) = N$.

Extreme Value Theorem (EVT)

If f is continuous on $[a, b]$, then f attains both an absolute maximum and an absolute minimum on $[a, b]$.

Rolle's Theorem

If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.

Mean Value Theorem (MVT)

If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

L'Hospital's Rule

For an indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$,

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

provided $g'(x) \neq 0$ near a and the new limit exists. May be applied more than once if the new form is still indeterminate. Rewrite other indeterminate forms ($0 \cdot \infty$, $\infty - \infty$, 1^∞, 0^0, ∞^0) as a quotient first.

Not Differentiable At

- A discontinuity (hole, jump, or vertical asymptote)
- A corner or cusp
- A vertical tangent

Critical Numbers

c is a critical number of f if c is in the domain of f and $f'(c) = 0$ or $f'(c)$ DNE.

First Derivative Test

At a critical number c, with f continuous at c:

- f' changes $-$ to $+$: local minimum
- f' changes $+$ to $-$: local maximum
- No sign change: neither

Second Derivative Test

If $f'(c) = 0$ and $f''(c)$ exists:

- $f''(c) > 0$: local minimum
- $f''(c) < 0$: local maximum
- $f''(c) = 0$: inconclusive

Point of Inflection

A point where $f(x)$ changes concavity. Candidates occur where $f''(x) = 0$ or $f''(x)$ DNE, but the concavity must actually change sign.

Candidates Test

For f continuous on a closed interval (so the EVT guarantees they exist), calculate the y-values at all endpoints and critical numbers. The lowest value is the absolute minimum and the highest is the absolute maximum.

Where vs. What

"Where" $= x$-value "What" $= y$-value

Proportionality

Direct: $a = kb$ Inverse: $a = \dfrac{k}{b}$ (k is constant)

Growth and Decay

If $\dfrac{dy}{dt} = ky$, then $y = Ce^{kt}$, where C is the initial value at $t = 0$ and k is constant.

$$k > 0: \text{ growth} \qquad k < 0: \text{ decay}$$

Motion Along a Line

If $s(t)$ is position, $v(t)$ is velocity, and $a(t)$ is acceleration:

- $v(t) = s'(t)$ and $a(t) = v'(t) = s''(t)$
- $\int a(t)\,dt = v(t) + C$ and $\int v(t)\,dt = s(t) + C$
- Speed $= |v(t)|$
- Displacement from $t = a$ to $t = b$: $s(b) - s(a) = \int_a^b v(t)\,dt$
- Total distance traveled: $\int_a^b |v(t)|\,dt$
- v and a have the same sign $\Rightarrow$ speeding up
- v and a have different signs $\Rightarrow$ slowing down

Indefinite Integrals

1. $\int 1\,dx = x + C$
2. $\int (u \pm v)\,dx = \int u\,dx \pm \int v\,dx$
3. $\int cv\,dx = c\int v\,dx$
4. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
5. $\int \frac{1}{x}\,dx = \ln|x| + C$
6. $\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C \quad (a \neq 0)$
7. $\int a^x\,dx = \frac{a^x}{\ln a} + C \quad (a > 0,\ a \neq 1)$
8. $\int \ln x\,dx = x\ln x - x + C$
9. $\int \log_a x\,dx = \frac{x\ln x - x}{\ln a} + C$
10. $\int u\,dv = uv - \int v\,du$
(Integration by Parts)
11. $\int (\sin u)u'\,dx = -\cos u + C$
12. $\int (\cos u)u'\,dx = \sin u + C$
13. $\int (\tan u)u'\,dx = \ln|\sec u| + C$
14. $\int (\cot u)u'\,dx = \ln|\sin u| + C$
15. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$
16. $\int \csc x\,dx = \ln|\csc x - \cot x| + C$
17. $\int (\sec^2 u)u'\,dx = \tan u + C$
18. $\int (\csc^2 u)u'\,dx = -\cot u + C$
19. $\int (\sec u\tan u)u'\,dx = \sec u + C$
20. $\int (\csc u\cot u)u'\,dx = -\csc u + C$
21. $\int_{(a\neq 0)} \frac{u'}{u^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$
22. $\int \frac{u'}{1+u^2}\,dx = \tan^{-1}(u) + C$
23. $\int_{(a>0)} \frac{u'}{\sqrt{a^2-u^2}}\,dx = \sin^{-1}\left(\frac{u}{a}\right) + C$
24. $\int \frac{u'}{\sqrt{1-u^2}}\,dx = \sin^{-1}(u) + C$
25. $\int \sinh x\,dx = \cosh x + C$
26. $\int \cosh x\,dx = \sinh x + C$
27. $\int \tanh x\,dx = \ln(\cosh x) + C$
28. $\int \operatorname{sech}^2 x\,dx = \tanh x + C$
29. $\int \operatorname{csch}^2 x\,dx = -\coth x + C$
30. $\int \operatorname{sech} x\tanh x\,dx = -\operatorname{sech} x + C$
31. $\int \operatorname{csch} x\coth x\,dx = -\operatorname{csch} x + C$

Definite Integrals

1. $\int_a^a f(x)\,dx = 0$
2. $\int_a^b c\,dx = c(b-a)$
3. $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
4. $\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$
5. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
6. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$
7. $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$
8. If $f(x) \geq 0$ on $[a,b]$ (with $a < b$), then $\int_a^b f(x)\,dx =$ area under $y = f(x)$
9. If $f(x) \leq 0$ on $[a,b]$, then $\int_a^b f(x)\,dx = -$(area between the curve and the x-axis)
10. $\int_{-a}^a f(x)\,dx = 0$ if f is **odd**
11. $\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$ if f is **even**
12. $\int_a^b |f(x)|\,dx =$ total area between the curve and the x-axis
13. $\int_a^b f'(x)g(x)\,dx = [f(x)g(x)]_a^b - \int_a^b f(x)g'(x)\,dx$
(Integration by Parts)

Definition of the Definite Integral

For f continuous on $[a,b]$, let $\Delta x = \dfrac{b-a}{n}$. Using right endpoints,

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} \left(\frac{b-a}{n}\right) f\left(a + \frac{b-a}{n}k\right).$$

Fundamental Theorem of Calculus

1. $\displaystyle\int_a^b f(x)\,dx = F(b) - F(a)$
   f continuous on $[a,b]$; F any antiderivative of f

2. $\displaystyle\int_a^b f'(x)\,dx = f(b) - f(a)$

3. $\displaystyle\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$
   f continuous

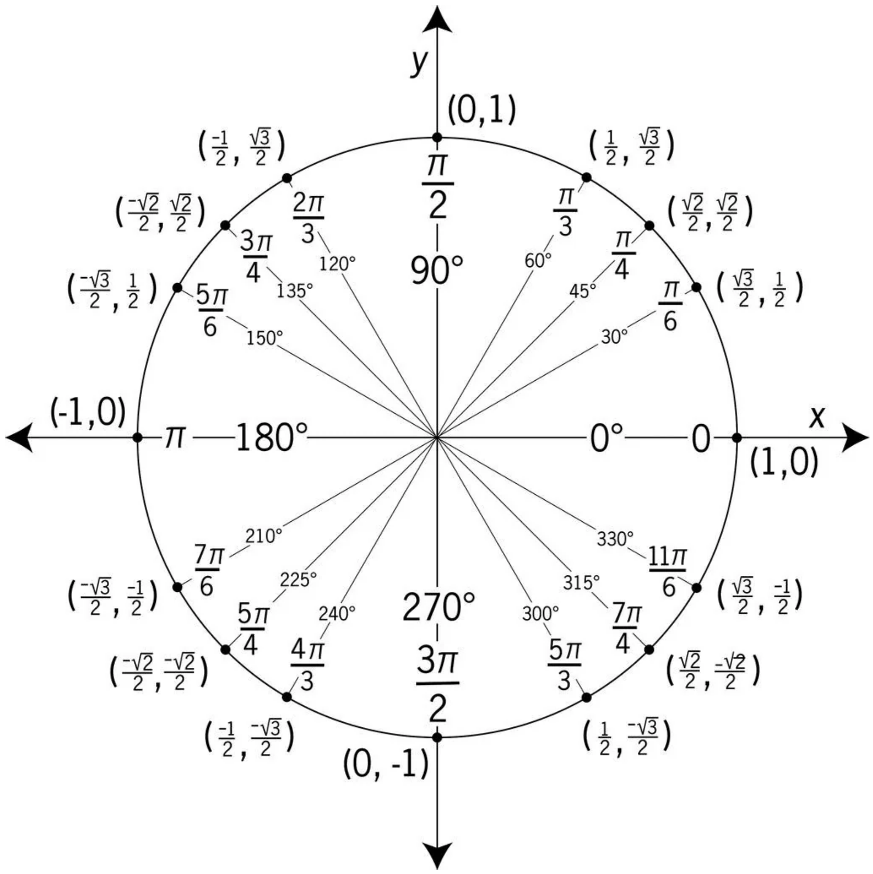
Variations (chain rule on the bounds)

4. $\displaystyle\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))\,g'(x)$

5. $\displaystyle\frac{d}{dx}\int_{h(x)}^{g(x)} f(t)\,dt = f(g(x))\,g'(x) - f(h(x))\,h'(x)$

Applications of the Definite Integral

- Average Value: $f_{\text{avg}} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$
- MVT for Integrals: if f is continuous on $[a,b]$, some $c \in [a,b]$ has $f(c) = f_{\text{avg}}$
- Area: $A = \displaystyle\int_a^b [f(x) - g(x)]\,dx \quad (f(x) \ge g(x) \text{ on } [a,b])$
- Horizontal Area: $A = \displaystyle\int_c^d [\,\text{right} - \text{left}\,]\,dy$
- Disk: $V = \pi\displaystyle\int_a^b [R(x)]^2\,dx$, $\quad R =$ distance from the axis to the outside curve
- Washer: $V = \pi\displaystyle\int_a^b \left([R(x)]^2 - [r(x)]^2\right)\,dx$, $\quad R =$ outside radius, $r =$ inside radius
- Shell: $V = 2\pi\displaystyle\int_a^b R(x)h(x)\,dx$, $\quad h(x) = f(x) - g(x)$
- Cross: $V = \displaystyle\int_a^b A(x)\,dx$
- Square: $V = \displaystyle\int_a^b S^2\,dx$, $\quad S = f(x) - g(x)$
- Semicircle: $V = \displaystyle\int_a^b \frac{1}{2}\pi r^2\,dx$, $\quad r = \dfrac{f(x) - g(x)}{2}$
- Rectangle: $V = \displaystyle\int_a^b wh\,dx$, $\quad w = f(x) - g(x)$, $\quad h$ given by the problem (often $h = kw$)
- Equilateral Tri: $V = \displaystyle\int_a^b \frac{\sqrt{3}}{4}S^2\,dx$, $\quad S = f(x) - g(x)$
- Isosceles *Right* Tri (legs $= S$): $V = \displaystyle\int_a^b \frac{1}{2}S^2\,dx$, $\quad S = f(x) - g(x)$



How to Read Math Notation

Limits

- $\lim_{x\to a} f(x) =$ "the limit of $f(x)$ as x approaches a"
- $\lim_{x\to\infty} f(x) =$ "as x goes to infinity"

Derivatives

- $\frac{d}{dx}[f(x)] =$ "derivative of $f(x)$ with respect to x"
- $f'(x) =$ "f prime of x"
- $\frac{dy}{dx} =$ "rate of change of y with respect to x"

Integrals

- $\int f(x)\,dx =$ "integral of $f(x)$ with respect to x"
- $dx =$ "with respect to x (tells the variable)"
- $\int_a^b f(x)\,dx =$ "integral from a to b (plug into $F(b)-F(a)$)"

Summation

- $\sum_{k=1}^{n} a_k =$ "sum from $k = 1$ to n of a_k"
- $k =$ index (the counting variable)
- $n =$ number of terms (ending value)

Function Composition

- $f(g(x)) =$ "plug $g(x)$ into f (inside first, then outside)"

Powers and Parentheses

- $(g(x))^2 =$ "square the whole function"
- $g^2(x) =$ "same as $(g(x))^2$"

Trig Formulas and Identities

Tangent and Cotangent Identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Reciprocal Identities

$$\csc\theta = \frac{1}{\sin\theta} \qquad \sin\theta = \frac{1}{\csc\theta}$$

$$\sec\theta = \frac{1}{\cos\theta} \qquad \cos\theta = \frac{1}{\sec\theta}$$

$$\cot\theta = \frac{1}{\tan\theta} \qquad \tan\theta = \frac{1}{\cot\theta}$$

Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

Even/Odd Formulas

$$\sin(-\theta) = -\sin\theta \qquad \csc(-\theta) = -\csc\theta$$

$$\cos(-\theta) = \cos\theta \qquad \sec(-\theta) = \sec\theta$$

$$\tan(-\theta) = -\tan\theta \qquad \cot(-\theta) = -\cot\theta$$

Periodic Formulas

If n is an integer.

$$\sin(\theta + 2\pi n) = \sin\theta \qquad \csc(\theta + 2\pi n) = \csc\theta$$

$$\cos(\theta + 2\pi n) = \cos\theta \qquad \sec(\theta + 2\pi n) = \sec\theta$$

$$\tan(\theta + \pi n) = \tan\theta \qquad \cot(\theta + \pi n) = \cot\theta$$

Double Angle Formulas

$$\sin(2\theta) = 2\sin\theta\cos\theta$$

$$\begin{aligned}\cos(2\theta) &= \cos^2\theta - \sin^2\theta \\ &= 2\cos^2\theta - 1 \\ &= 1 - 2\sin^2\theta\end{aligned}$$

$$\tan(2\theta) = \frac{2\tan\theta}{1 - \tan^2\theta}$$

Degrees to Radians Formulas

If x is an angle in degrees and t is an angle in radians then

$$\frac{\pi}{180} = \frac{t}{x} \quad \Rightarrow \quad t = \frac{\pi x}{180} \quad \text{and} \quad x = \frac{180t}{\pi}$$

$$\tan(A/2) = \frac{\sin A}{1 + \cos A}$$

$$\tan(A/2) = \frac{1 - \cos A}{\sin A}$$

Half Angle Formulas

$$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1 - \cos\theta}{2}}$$

$$\cos\frac{\theta}{2} = \pm\sqrt{\frac{1 + \cos\theta}{2}}$$

$$\tan\frac{\theta}{2} = \pm\sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}}$$

Power-Reducing

$$\sin^2\theta = \frac{1}{2}(1 - \cos(2\theta))$$

$$\cos^2\theta = \frac{1}{2}(1 + \cos(2\theta))$$

$$\tan^2\theta = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}$$

Sum and Difference Formulas

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

Product to Sum Formulas

$$\sin\alpha\sin\beta = \frac{1}{2}\left[\cos(\alpha - \beta) - \cos(\alpha + \beta)\right]$$

$$\cos\alpha\cos\beta = \frac{1}{2}\left[\cos(\alpha - \beta) + \cos(\alpha + \beta)\right]$$

$$\sin\alpha\cos\beta = \frac{1}{2}\left[\sin(\alpha + \beta) + \sin(\alpha - \beta)\right]$$

$$\cos\alpha\sin\beta = \frac{1}{2}\left[\sin(\alpha + \beta) - \sin(\alpha - \beta)\right]$$

Sum to Product Formulas

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha + \beta}{2}\right)\cos\left(\frac{\alpha - \beta}{2}\right)$$

$$\sin\alpha - \sin\beta = 2\cos\left(\frac{\alpha + \beta}{2}\right)\sin\left(\frac{\alpha - \beta}{2}\right)$$

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha + \beta}{2}\right)\cos\left(\frac{\alpha - \beta}{2}\right)$$

$$\cos\alpha - \cos\beta = -2\sin\left(\frac{\alpha + \beta}{2}\right)\sin\left(\frac{\alpha - \beta}{2}\right)$$

Cofunction Formulas

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta \qquad \cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$$

$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec\theta \qquad \sec\left(\frac{\pi}{2} - \theta\right) = \csc\theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta \qquad \cot\left(\frac{\pi}{2} - \theta\right) = \tan\theta$$

BC Topics: Series, Parametric & Polar, Differential Equations, Approximation

Series Convergence Tests

1. **nth-Term (Divergence) Test.** If $\lim_{n\to\infty} a_n \neq 0$ or DNE, then $\sum a_n$ diverges. If $\lim a_n = 0$, the test is *inconclusive*.
2. **Geometric.** $\sum_{n=0}^{\infty} ar^n$ converges to $\dfrac{a}{1-r}$ if $|r| < 1$; diverges if $|r| \geq 1$.
3. **p-Series.** $\sum \dfrac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$ ($p = 1$ is the harmonic series).
4. **Integral Test.** If f is positive, continuous, and decreasing on $[k, \infty)$ with $f(n) = a_n$, then $\sum a_n$ and $\int_k^\infty f(x)\,dx$ both converge or both diverge. (The integral is not the sum.)
5. **Direct Comparison.** For $0 \leq a_n \leq b_n$: $\sum b_n$ converges $\Rightarrow \sum a_n$ converges; $\sum a_n$ diverges $\Rightarrow \sum b_n$ diverges.
6. **Limit Comparison.** If $a_n, b_n > 0$ and $\lim \dfrac{a_n}{b_n} = L$ with $0 < L < \infty$, then both converge or both diverge.
7. **Alternating Series Test.** $\sum (-1)^n b_n$ with $b_n > 0$ converges if b_n is eventually decreasing *and* $\lim b_n = 0$.
8. **Ratio Test.** $L = \lim \left|\dfrac{a_{n+1}}{a_n}\right|$: $L < 1$ converges absolutely, $L > 1$ (or ∞) diverges, $L = 1$ inconclusive.
9. **Root Test.** $L = \lim \sqrt[n]{|a_n|}$: same conclusions as the ratio test.
10. **Absolute vs. Conditional.** $\sum |a_n|$ converges $\Rightarrow$ *absolute* convergence. $\sum a_n$ converges but $\sum |a_n|$ diverges $\Rightarrow$ *conditional*.

Taylor and Maclaurin Series

Taylor series centered at $x = a$ (Maclaurin is the case $a = 0$), then the degree-n Taylor polynomial P_n:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

Common Maclaurin Series

- $e^x = \displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$ (all x)
- $\sin x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ (all x)
- $\cos x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$ (all x)
- $\dfrac{1}{1-x} = \displaystyle\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$ ($|x| < 1$)
- $\ln(1+x) = \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ ($-1 < x \leq 1$)
- $\tan^{-1} x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ ($|x| \leq 1$)

Radius and Interval of Convergence

Apply the ratio test to $\sum c_n (x-a)^n$ and solve $L < 1$: converges for $|x - a| < R$, diverges for $|x - a| > R$. Test the endpoints $x = a \pm R$ *separately* by substitution. $R = 0$ (only $x = a$) and $R = \infty$ (all x) are possible.

Error Bounds

Alternating series (when the AST conditions hold): the error is at most the first omitted term,

$$|S - S_n| \leq b_{n+1}.$$

Lagrange error for P_n centered at a:

$$|f(x) - P_n(x)| \leq \frac{\max\left|f^{(n+1)}(z)\right|}{(n+1)!}|x-a|^{n+1}$$

where z is between a and x.

Parametric Equations

- $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$ $(dx/dt \neq 0)$
- $\dfrac{d^2y}{dx^2} = \dfrac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$ (not the quotient of 2nd derivatives)
- Velocity $\langle x'(t), y'(t)\rangle$; acceleration $\langle x''(t), y''(t)\rangle$
- Speed $= \sqrt{[x'(t)]^2 + [y'(t)]^2}$
- Arc length / total distance: $L = \displaystyle\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$
- $x(t) = x(t_0) + \displaystyle\int_{t_0}^{t} x'(\tau)\,d\tau$ (same for y)

Polar

- $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$, $\tan\theta = \dfrac{y}{x}$
- Area: $A = \dfrac{1}{2}\displaystyle\int_\alpha^\beta r^2\,d\theta$
- Between two curves: $A = \dfrac{1}{2}\displaystyle\int_\alpha^\beta \left(r_{\text{outer}}^2 - r_{\text{inner}}^2\right)d\theta$
- Slope: with $x = r(\theta)\cos\theta$, $y = r(\theta)\sin\theta$, use $\dfrac{dy}{dx} = \dfrac{dy/d\theta}{dx/d\theta}$

Arc Length (Cartesian)

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx \quad \text{or} \quad L = \int_c^d \sqrt{1 + [g'(y)]^2}\,dy$$

Differential Equations

- **Separable:** $\dfrac{dy}{dx} = f(x)g(y) \Rightarrow \displaystyle\int \frac{dy}{g(y)} = \int f(x)\,dx$
- **Slope field:** at each (x, y) draw a short segment of slope dy/dx.
- **Euler's method:** with step size h, $x_{n+1} = x_n + h$, $y_{n+1} = y_n + h\,f(x_n, y_n)$
- **Logistic:** $\dfrac{dP}{dt} = kP\left(1 - \dfrac{P}{L}\right)$, carrying capacity L, $\lim_{t\to\infty} P = L$, fastest growth at $P = \dfrac{L}{2}$.
  Solution: $P(t) = \dfrac{L}{1 + Ae^{-kt}}$, $A = \dfrac{L - P_0}{P_0}$

Improper Integrals

- $\displaystyle\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$
- $\displaystyle\int_{-\infty}^\infty f\,dx = \int_{-\infty}^c f\,dx + \int_c^\infty f\,dx$ (both must converge)
- Infinite discontinuity at b: $\displaystyle\int_a^b f\,dx = \lim_{t\to b^-}\int_a^t f\,dx$; converges iff the limit is finite.

Partial Fractions

Requires $\deg P < \deg Q$ (otherwise divide first). $\dfrac{P(x)}{(x-r_1)(x-r_2)} = \dfrac{A}{x-r_1} + \dfrac{B}{x-r_2}$; repeated factor $(x-r)^2$ gives $\dfrac{A}{x-r} + \dfrac{B}{(x-r)^2}$.

Riemann Sums and the Trapezoidal Rule

With $\Delta x = \dfrac{b-a}{n}$ and $x_i = a + i\,\Delta x$:

- Left: $\Delta x\,[f(x_0) + f(x_1) + \cdots + f(x_{n-1})]$
- Right: $\Delta x\,[f(x_1) + f(x_2) + \cdots + f(x_n)]$
- Midpoint: $\Delta x\left[f\left(\frac{x_0+x_1}{2}\right) + \cdots + f\left(\frac{x_{n-1}+x_n}{2}\right)\right]$
- Trapezoid: $\dfrac{\Delta x}{2}\,[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)]$
- Unequal widths: add $\frac{1}{2}$(width)(left + right height) per strip.
- f increasing: left under, right over. f concave up: trapezoid over, midpoint under.

Linearization, Differentials, Related Rates

- $L(x) = f(a) + f'(a)(x-a)$, and $f(x) \approx L(x)$ for x near a
- $dy = f'(x)\,dx$, $\Delta y \approx dy$
- Concave up at a: tangent lies below the curve, so L *under*estimates. Concave down: L *over*estimates.
- Related rates: differentiate the relating equation with respect to t, substitute known values *last*, solve for the unknown rate.